

# OSI & TCP/IP Network Models

SOC Analyst Technical Reference Notes

## OSI Model - 7 Layers

### What is OSI Model?

The OSI Model (Open Systems Interconnection) is a 7-layer framework that explains how data travels from one system to another. Think of it as a step-by-step journey of data.

### 7 Layers of OSI Model:

- **Layer 7:** Application (Example: HTTP, FTP)
- **Layer 6:** Presentation (Example: Encryption)
- **Layer 5:** Session (Example: Session management)
- **Layer 4:** Transport (Example: TCP/UDP)
- **Layer 3:** Network (Example: IP)
- **Layer 2:** Data Link (Example: MAC)
- **Layer 1:** Physical (Example: Cables)

### 7. Application Layer

The application layer is the topmost layer of the OSI model and is closest to the end user. This layer provides the interface for users and software applications to interact with the network. It handles high-level functions like file transfers, email, and web browsing.

- **What it does:** User interacts here.
- **Examples:** Web browsing, Email, File transfer.
- **Protocols:** HTTP/HTTPS, FTP, SMTP.

**SOC Perspective:** Phishing attacks, Malicious URLs, Suspicious API calls.

### 6. Presentation Layer

The presentation layer is responsible for the translation, encryption, and compression of data. It ensures that data sent from the application layer is in a format that can be understood by the receiving system. This layer often handles tasks like data formatting, character encoding (e.g., ASCII to Unicode), and data encryption.

- **What it does:** Data formatting, Encryption / Decryption, Encoding.
- **Examples:** SSL/TLS.

**SOC Perspective:** Encryption analysis, SSL stripping (MITM).

## 5. Session Layer

The session layer establishes, manages, and terminates sessions between two communicating devices. It ensures that data exchange is synchronized and allows applications to communicate effectively without interference.

- **What it does:** Maintains session between systems.
- **Example:** Login session, Session cookies.

**SOC Perspective:** Session hijacking, Token reuse.

## 4. Transport Layer

The transport layer ensures the reliable delivery of data between two devices across a network. It segments the data received from the upper layers and provides error detection and correction. Additionally, it controls flow to prevent congestion and guarantees that data is delivered in the correct order.

- **What it does:** End-to-end communication, Reliability.
- **Protocols:** TCP (reliable), UDP (fast, unreliable).

**SOC Perspective:** Port scanning, SYN flood attacks, Suspicious ports.

## 3. Network Layer

The network layer is responsible for determining the best path to send data across multiple networks. This layer handles logical addressing and routing, enabling data to travel from one network to another, regardless of the physical infrastructure. Routers operate at this layer, forwarding data between different networks based on IP addresses.

- **What it does:** Routing, IP addressing.
- **Protocol:** IP (IPv4/IPv6).

**SOC Perspective:** IP tracking, Geolocation, Suspicious external connections.

## 2. Data Link Layer

The data link layer sits above the physical layer and is responsible for the reliable transfer of data across a physical network link. This layer ensures that data is error-free and properly framed for transmission. It also manages how devices on the same local network communicate with each other.

- **What it does:** MAC address communication, Local network delivery.

**SOC Perspective:** ARP spoofing (MITM), MAC anomalies.

## 1. Physical Layer

The physical layer is the lowest layer of the OSI model, and it deals with the actual transmission of raw binary data (1s and 0s) over a physical medium, such as cables, fiber optics, or wireless signals. It defines the

hardware elements involved in data transfer, such as network interfaces, cables, switches, and the electrical signals used to carry the data.

- **What it does:** Hardware transmission, Electrical signals.

**SOC Perspective:** Rarely used directly, but important in hardware attacks.

## How does communication happen in the OSI Model

Communication in the OSI Model follows a systematic process called **Encapsulation and Decapsulation**:

1. **Sender's Side:** Data is generated at the application layer and moves downward through the layers. Each layer adds its own header, and sometimes a footer, encapsulating the data with layer-specific information (e.g., addressing, error checks).
2. **Transmission:** The encapsulated data (now called a packet) is transmitted over the physical medium to the receiving device.
3. **Receiver's Side:** The process is reversed as data moves upward through the OSI layers. Each layer removes its corresponding header/footer (decapsulation) until the original data is presented to the receiving application.

*\*Note: This structured approach ensures data integrity and reliability, even over complex networks.\**

## TCP/IP Model (Real-World Model)

**What is TCP/IP Model?**

- Used in real networks (Internet).
- Simplified version of OSI.

### TCP/IP Layers Mapping to OSI:

- **Application Layer** → Maps to OSI 7, 6, 5
- **Transport Layer** → Maps to OSI 4
- **Internet Layer** → Maps to OSI 3
- **Network Access Layer** → Maps to OSI 2, 1

### Detail of TCP/IP Layers:

1. **Application Layer:**  
Includes: HTTP, HTTPS, DNS, FTP
  - **SOC View:** Phishing, DNS attacks, API abuse
2. **Transport Layer:**  
Protocols: TCP, UDP
  - **SOC View:** Port scanning, SYN flood, Traffic anomalies

### 3. Internet Layer:

Protocol: IP

- SOC View: Suspicious IP, External communication, Botnet traffic

### 4. Network Access Layer:

Includes: MAC, ARP, Ethernet

- SOC View: ARP spoofing, Internal attacks

## OSI vs TCP/IP (Important for Interviews)

| Feature    | OSI        | TCP/IP    |
|------------|------------|-----------|
| Layers     | 7          | 4         |
| Use        | Conceptual | Practical |
| Real usage | Less       | High      |

## SOC Example & Analyst Mindset

**Alert Case Study:** Suspicious login via HTTPS from unknown IP

- **Application Layer** → HTTP login
- **Transport Layer** → Port 443
- **Network Layer** → External IP
- **Data Link Layer** → Internal network

Whenever you look at an individual log, break it down using the **SOC Analyst Thinking Matrix**:

- Port issue → **Transport Layer**
- IP issue → **Network Layer**
- DNS issue → **Application Layer**
- ARP issue → **Data Link Layer**